

**THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
SHINYANGA REGION
FORM FOUR MOCK EXAMINATION 2024
PHYSICS 1**

Code: 031/1

Time: 3:00 Hours

INSTRUCTIONS

- ✓ This paper consists of three sections A,B and C with total of eleven (11) questions
- ✓ Answer **all** questions in section A and B and two (2) questions from section C
- ✓ Communication devices and unauthorized materials are not allowed in examination room
- ✓ Non-programmable calculator and mathematical table may be used.
- ✓ Write your examination number on every page of your answer sheet (s) provided
- ✓ Where necessary the following constants may be used;
 - i Acceleration due to gravity, $g=10\text{m/s}^2$
 - ii $\text{Pie}=3.14$
 - iii Linear expansivity of steel $=0.00001\text{per } ^\circ\text{C}$
 - iv Linear expansivity of brass $=0.000019 \text{ per } ^\circ\text{C}$

FOR EXAMINER'S USE ONLY		
QUESTION	SCORE	EXAMINER'S INITIAL
01		
02		
3		
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9		
10		
11		
Total		
Checker's initial		

Gepam Science Hub

SECTION A. (16Marks)

Answer **all** questions in this section.

1. For each of the item (i-x) choose the most correct answer from among the given alternatives and write its letter beside the item number in the answer sheet provided.

(10 marks)

- i Light wave differ from sound wave because;
A. Light is an electromagnetic wave
B. Light wave are long and sound wave are short
C. Interference is obtained with light waves but not with sound waves []
D. The speed of light is independent with medium in which it travel
E. Sound waves do not travel in water.
- ii Mercury form spherical drops when split on a wooden bench because;
A. Its velocity is high
B. It has high cohesive force
C. It has high relative density []
D. It has low surface tension
E. It has high adhesive force
- iii Hydrometer is an instrument for measuring density or relative density of a liquid. What are you supposed to do in order to increase its sensitivity?.
A. Increasing the size of the large bulb
B. Making the stem narrower
C. Reducing the lead shots in the weighted bulb []
D. Increasing the length of the stem
E. Reducing the length of stem and lead shots
- iv Which statement is true about ball falling freely from height of 10m?.
A. Its potential energy increase but kinetic energy decrease
B. It potential energy is equal to kinetic energy
C. Its potential energy is zero and kinetic energy is maximum []
D. Its potential energy decrease and kinetic energy increase
E. Kinetic energy and potential energy are maximum
- v The following statements about magnetic lines of force are correct **except**;
A. Always form close loops
B. Start at north pole and end at south pole []
C. Cross one another
D. Stronger where the lines are closer together
E. Pass through all material magnetic and non-magnetic material
- vi The half life of radioactive material is 12hours. What fraction of the element will have disintegrated in 72 hours?
A. 1/64 B. 1/16 C. 1/32 D. 1/8 E. 1/72 []
- vii In a single rope 4-pulley system, the mechanical advantage is less than 4 because;
A. The effort may vary
B. The load is raised
C. The upper pulley do not move
D. The friction acts on the pulley []
E. The weight of the lower pulley can be neglected
- viii A car moving at steady speed has friction force on its surface whose size depends on;
A. The speed and surface are
B. Its speed only
C. Its surface area only []
D. Its weight only
E. Its wheel speed only
- ix A study that deals with constellation, solar system and cosmos is referred to as;
A. Electromagnetism B. Structure and properties of matter C. Light []
D. Astronomy E. Geophysics

- x The cost of electricity for a 2000W electric fire used for 10.30 hours at the rate of shs 8 per Kwh is;
 A. Shs 160 B. shs80 C. shs 168 D. shs 1600 E. shs3200 []

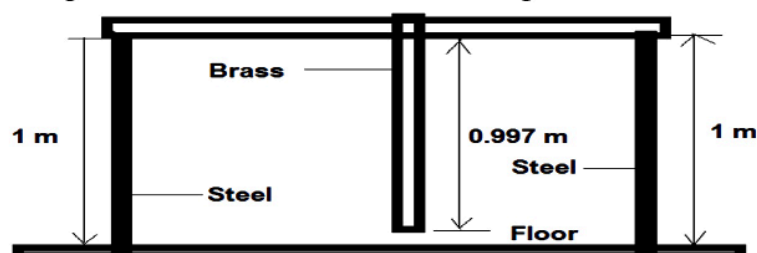
2. Match the items in **List A** with the responses in **List B** by writing the letter of the corresponding response beside the item number in the answer sheet provided. **(6 marks)**

LIST A	LIST B
i. Longitudinal wave	A. $\frac{\sin i}{\sin r}$
ii. Reverberation	B. Refractive index
iii. Snell's law	C. Water waves
iv. Critical angle	D. Sound wave
v. Principle of superposition	E. Angle of incidence for which angle of refraction is 90°
vi. Echo	F. Rise and fall of sound of two vibrating objects
	G. The resultant of displacement at any point is equal to the sum of displacements of different waves
	H. The multiple reflection of sound waves
	I. Angle of reflection for which angle of incidence is 90°
	J. Reflected sound.

SECTION B (54marks)

Answer **all** questions in this section

3. a) Explain why it is easy to open nut by using spanner with long arm than that of short arm **(3marks)**
 b) A metal beam AB of mass 50kg is supported at its ends. The beam carries a mass of 200kg at a distance of 0.75m from end A. If the beam is 2m long, determine the thrust at supports A and B. **(6 marks)**
4. a) Why convex mirror is used as side mirrors?. **(3 marks)**
 b) An object 10 cm tall stands vertically on the principal axis of a convex lens of focal length 10 cm and at a distance of 17 cm from the lens. By means of accurate graphical construction find the position, size and nature of the image formed **(6 marks).**
5. a) Why bi-metallic strip tends to bend when heated under high temperature or cooled under low temperature?. **(3 marks)**
 b) A brass rod of length of 0.997 m at 20°C is hung from a steel framework with height of 1m at 20°C as shown in the figure below



At what temperature would the brass rod just touch the floor?

- (6 marks)**
6. a) Explain why people at high altitude of a mountain suffers from nose bleeding. **(3 marks)**
 b) With aid of clear diagram explain how pressure increase with depth. **(6 marks)**
7. a) Half-life of a certain radioactive material is 1200years. What does this statement mean?. **(3 marks)**
 b) The half-life for the Beta decay of Thorium-234 is 24 days. Department of physics has brought sample of Thorium from German. On the day of dispatch its activity was 4×10^5 disintegrations per second.
 i. What was the activity of the source 72 days latter?. **(3 marks)**

- ii. What safety precaution should the supplier take to ensure that non of dock workers should be harmed? **(3 marks)**
8. a) Briefly explain the meaning of the following terms as used on earthquake:
 (i) Hypocentre
 (ii) Epicentre **(4 marks)**
- b) You are working at village where people are mainly using charcoal as a source of energy. Explain to them effect that might arise to the environment as result of prolonged use of charcoal.(5 points) **(5marks)**

SECTION C. (30 marks)

Answer two (2) questions from this section

9. a) Explain briefly why sound produced in hall with many people is heard more clearly than when the hall has few people?. **(3 marks)**
- b) The frequency obtained from a plucked string is 400Hz when the tension is 2N. Calculate
 i. The frequency when Tension is increased to 8N **(3 marks)**
 ii. Tension needed to produce a note of frequency of 600Hz **(3 marks)**
- c) In a closed pipe, the first resonance is at 23 cm and second at 73 cm. Determine the wavelength of the sound and the end correction of the pipe. **(6 marks)**
10. a) With aid of clear diagram explain mechanism of fuse **(5 marks)**
- b) A circuit in a house is protected by a 10A fuse. The circuit is connected to 240V main. The following appliances are connected to the circuit;

<i>Appliance</i>	<i>Power rating</i>
Bulb 1	100W
Bulb 2	75W
TV	300W
Heater	1500W

- Determine whether the fuse will blow on or off if all appliances are turned on. **(5 marks)**
- c) i) State how short circuit occurs in a house? **(2.5 marks)**
 ii) Mention two causes of electrical short-circuit **(2.5 marks)**
11. a) i) State the laws of electromagnetic induction. **(2 marks)**
 ii) Mention two advantages of a.c. generators over d.c. generators. **(3 marks)**
- b) i) Briefly explain why the core of a transformer is made of thin layers of metal insulated from one another. **(2.5 marks)**
 ii) Describe the structure of a step-up transformer and state how it works. **(2.5 marks)**
- c) A step-up transformer has 10000 turns in the secondary coil and 100 turns in the primary coil. current of 5A flow in the primary circuit when connected to a 12V supply.
 i) Calculate voltage across the secondary coil **(2 marks)**
 ii) If the transformer has efficiency of 90% what is the current in the secondary coil? **(3 marks)**